SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA51

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 5 **Total Marks:50**

Annexure A

APPLICATION- BASED QUESTIONS

Code of Conduct:

B. Nandha Kishore-192324159 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

APPLICATION- BASED QUESTIONS

1. Scenario: Event Communication Security
A student organization is preparing to share confidential event details with its core members using a simple classical encryption method. During testing, a message "MEET AT NOON" is transformed into an encoded version using a fixed shift pattern.

a. Reconstruct how the encrypted message might appear if a shift of 4 positions is applied.
b. Explain how an authorized member, aware of the encoding rule, would retrieve the original message.

2. Scenario: Intercepted Suspicious Message
A cybersecurity intern intercepts the coded message: "GSRH RH Z HVXIVG" during network monitoring. The structure of the text suggests a simple substitution mechanism rather than complex encryption.

a. Deduce the most likely original message.
b. Justify your answer by analyzing repeating patterns, letter structures, or known cipher characteristics.

3. Scenario: Secure Messaging System
A company deploys a keyword-based encryption technique for internal communication. An employee sends the word "HELLO" using the keyword "KEY" to ensure confidentiality.

a. Demonstrate how the encryption process transforms the message step-by-step.
b. Explain how the repeating keyword influences each character of the ciphertext.

4. Scenario: Suspicious Digital Asset Investigation
During a digital forensic audit, investigators suspect that a company logo file may contain hidden confidential information embedded within it. The file appears normal upon casual inspection.

a. Analyze how investigators could detect the presence of hidden data within the image.
b. Recommend two techniques that could strengthen the security of such hidden communications against detection or extraction.

5. Scenario: Legacy Encryption in Corporate Communication
A legacy system in a company still uses a transposition-based encryption method with three levels of arrangement. A received message appears as: "WECRLTEERDSOEEFEAOCAIVDEN".

a. Determine the most probable original message by reversing the encryption process.
b. Critically evaluate why such a method is vulnerable in modern communication systems.
c. Propose one practical enhancement to improve the confidentiality of messages in this scenario.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools

Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

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(1) Event communication security – caeser cipher.

a. Encryption using a shift of 4 positions.

Plaintext: MEET AT NOON

Each letter is shifed Four positions Forward in the alphabet

~~M → Q~~

$C = (P + 4) \bmod 26.$

Plaintext	Shift operation	Cipher text
M	M → N → O → P → Q	Q
E	E → F → G → H → I	I
E	E → F → G → H → I	I
T	T → U → V → W → X	X
A	A → B → C → D → E	E
T	T → U → V → W → X	X
N	N → O → P → Q → R	R
O	O → P → Q → R → S	S
O	O → P → Q → R → S	S
N	N → O → P → Q → R	R

Ciphertext : QIIX EX RSSR.

b. Explain how an authorized member retrieves the original message.

⇒ The receiver obtains the ciphertext QIIx Ex RSSR.

⇒ The receiver knows that the encryption shift is 4

⇒ each cipher text letter is shifted four positions back ward

Decryption Formula

$$P = (C - 4) \bmod 26$$

Step 4: The receiver obtains the original message.

Decrypted message : MEET AT NOON.

2

Intercepted suspicious message

(a) Deduce the most likely original message.

Given cipher text : GSRH RH Z HVXIVG

Write the alphabet mapping :-

plain alphabet	A	B	C	D	E	F	G	H	I	J	K	L	M
Atbash alphabet	Z	Y	X	W	V	U	T	S	R	Q	P	O	N

The remaining letters follow the same reverse-alphabet rule.

Secure Messaging System

(3)

(a) Demonstrate the encryption of "HELLO" using the key word "KEY"

Write the plain text and keyword.

plaintext : HELLO

Key word : KEY.

Repeat the keyword until it matches the length of the plaintext

Repeated keyword : KEYKE

Assign numerical values to the letters

Use A=0 B=1 Z=25

Letter	H	E	L	L	O
plaintext value	7	4	11	11	14
keyword letter	K	E	Y	K	E
keyword value	10	4	24	10	4

APPLY the encryption Formula :-

$$C = (P+K) \bmod 26.$$

=> H+K (7+10) mod 26 = 17 → R.

=> E+E (4+4) mod 26 = 8 → I

=> L+Y (11+24) mod 26 = 9 → J

=> L+K (11+10) mod 26 = 21 → V

=> O+E (14+4) mod 26 = 18 → S

Substitute each ciphertext letter.

Ciphertext G S R H R H Z H V x I V G

Plaintext T H I S I S A S E C R E T

Restore The Spaces

Cipher text:- GSRH RH Z HVxIVG

plain text:- THIS IS A SECRET

(b) Justify The answer using letter patterns and cipher characteristics

⇒ The cipher text contains four words with lengths 4, 2, 1, and 6

⇒ The Atbash cipher preserves word lengths and spaces because it replaces each letter without rearranging The text

⇒ Applying The reverse-alphabet maping converts GSRH into THIS RH int IS and Z int A

⇒ The complete decrypted sentence is meaningful english "THIS IS A SECRET"

Ixlrite The ciphertext.

plaintext : HELLO

Repeated keyword : KEYKE

ciphertext : RIJVS

(b) Explain How The repeating keyword influences the cipher text

=> The keyword KEY is repeated to produce KEYKE

-> Each keyword letter determines The shift applied to The corresponding plain text letter.

=> Different keyword letters produce difficult shifts For example it shifts H to R while E shifts E to I

=> The resulting ciphertext is RIJVS

(4)

Suspicious Digital Asset investigation

(a) Analyze how investigators can detect hidden data within an image.

Examine File properties

=> Investigators inspect The image format File size dimensions and metadata to identify unusal characteristics

Analyze metadata :-

They examine EXIF information comments and other metadata fields For suspicious or unexpected

(5) Legacy Encryption in corporate communication.

(a) Determine the original message by reversing the three level transposition process.

Given cipher text:-

WECRLTEERDSOEEFEAOCAIIVDEN

Identify the encryption technique.

A three-rail Fence cipher is a possible interpretation of a transposition cipher with three levels

Count the Ciphertext characters

The cipher text contains 24 characters

Identify the Zigzag pattern.

For three rails the letters follow this repeating rail patterns.

Rail 1 → Rail 2 → Rail 3 → Rail 2 → Rail 1

Reconstruct the original message.

marking the rail positions and rearranging the cipher letters according to this pattern produces the following result for the supplied cipher text

(b) Critically evaluate the Vulonelabilities of this method

=> A transposition cipher changes the positions of a character but does not change the characters themselves.

-> The original letter frequencies remain unchanged. which can provide useful clues to an attacker.

-> A Three rail arrangement has a small number of position patterns making trial and error attacks practical.

=> it doesnot independently provide message integrity (or) authentication.

(c) Propose one practical enhancement

=> Replace the legacy transposition cipher with AES-256 encryption.

=> Generate and securely store a strong ~~Stre~~ Secret key.

=> encrypt the corporate message using an authenticated encryption mode such as AES-GCM

=> share the key securely with authorized recipients.

=> The receiver uses the correct key to decrypt the message and verify its integrity.